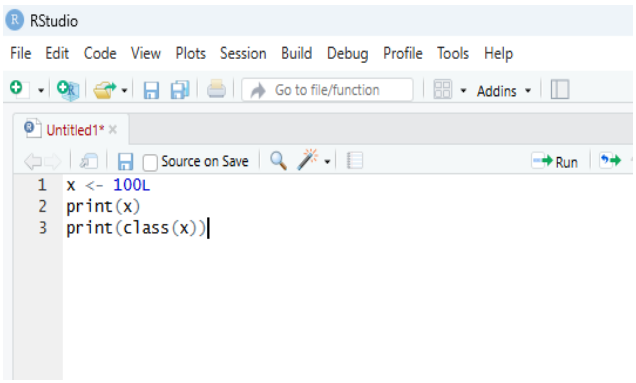
NAME : YUGITHA B
REG NO : 192424102

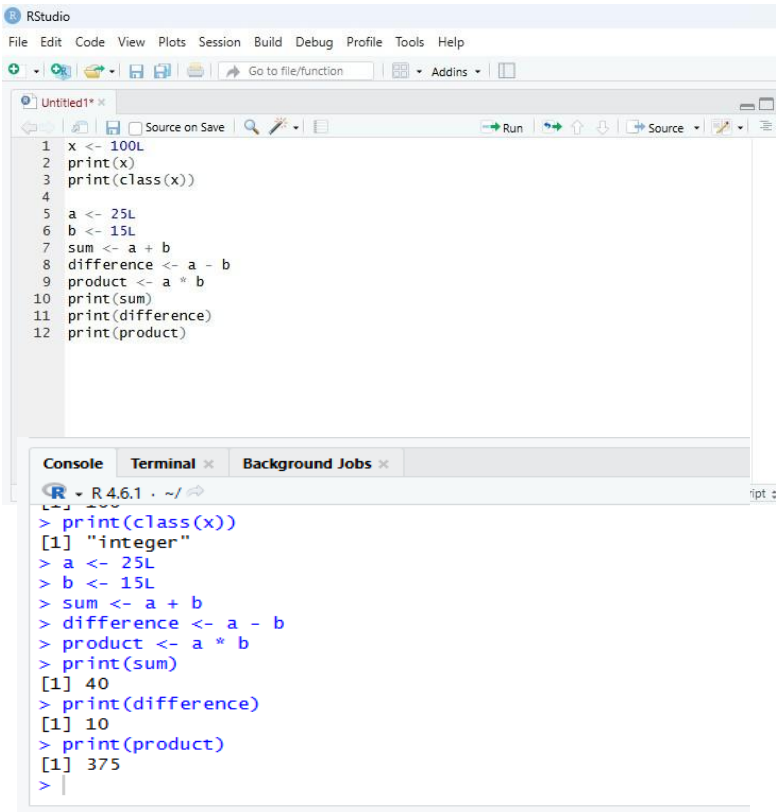
Integer Data Type

1. Create an integer variable with value 100 and display its data type

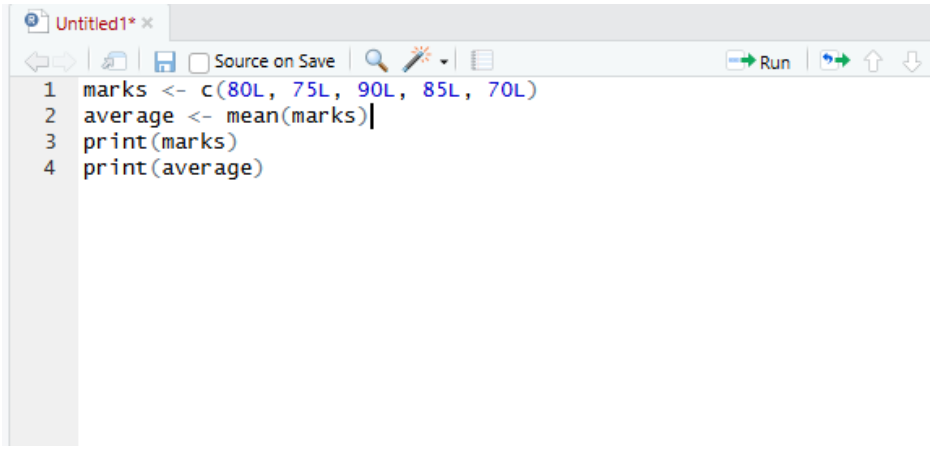


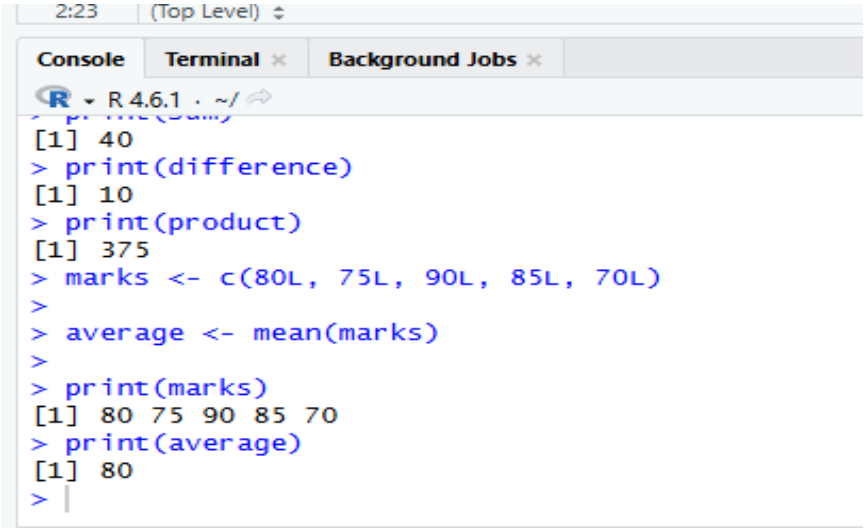

```
> x <- 100L
> print(x)
[1] 100
> print(class(x))
[1] "integer"
>
```

2. Create two integer variables and find sum, difference, and product



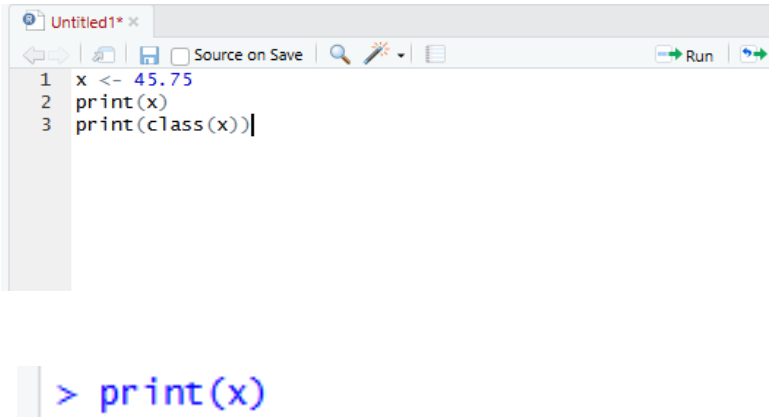
3. Store marks of 5 students as integers and calculate average

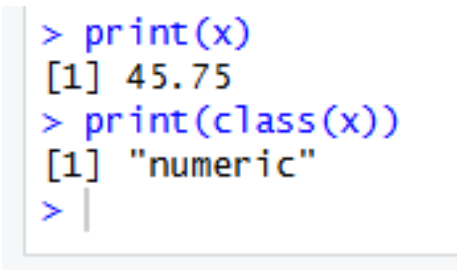




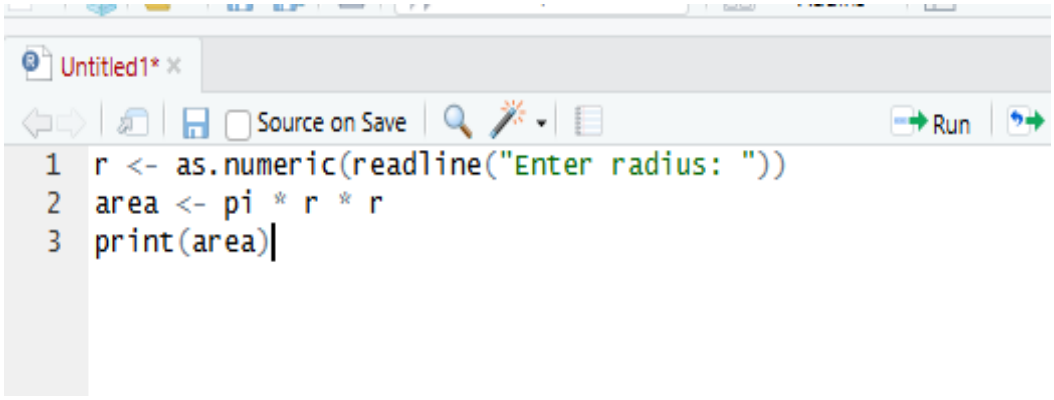
Numeric Data Type

4. Create a numeric variable with value 45.75 and check its class



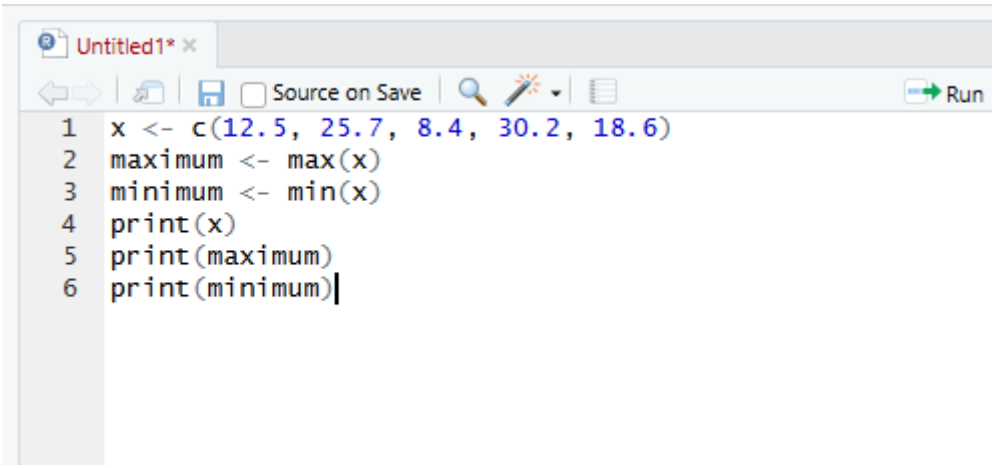


5. Calculate area of a circle using radius entered by user




```
> r <- as.numeric(readline("Enter radius: "))
Enter radius: 5
> area <- pi * r * r
> print(area)
[1] 78.53982
>
```

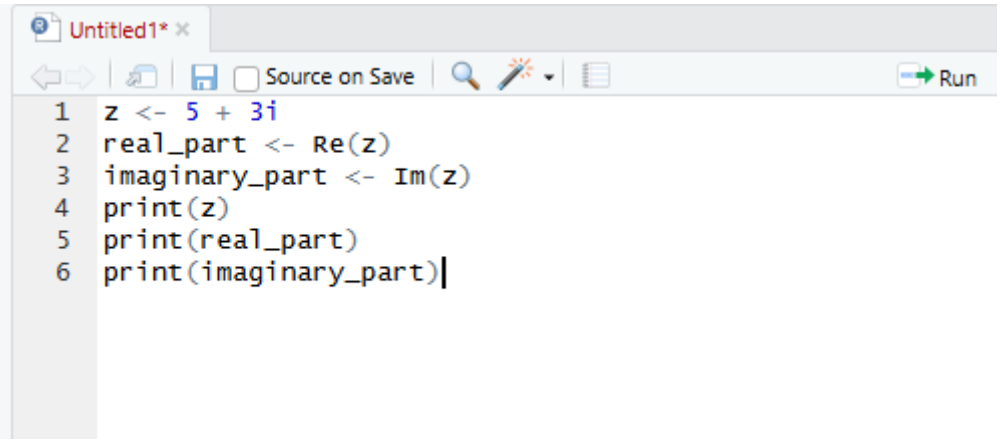
6. Create a numeric vector with five decimal values and find maximum and minimum




```
> x <- c(12.5, 25.7, 8.4, 30.2, 18.6)
> maximum <- max(x)
> minimum <- min(x)
> print(x)
[1] 12.5 25.7  8.4 30.2 18.6
> print(maximum)
[1] 30.2
> print(minimum)
[1] 8.4
>
```

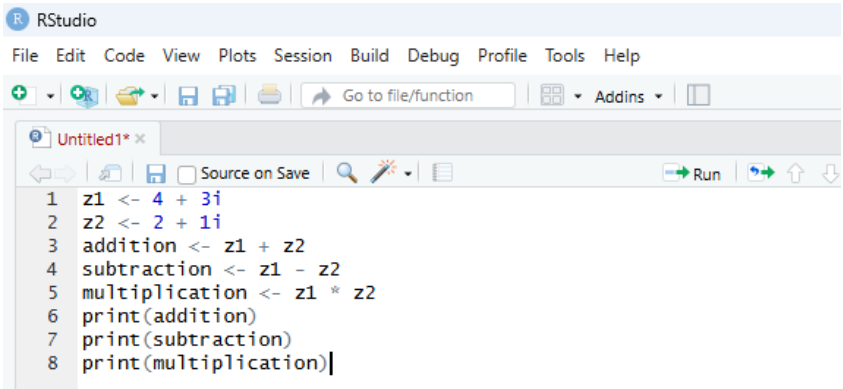
Complex Data Type

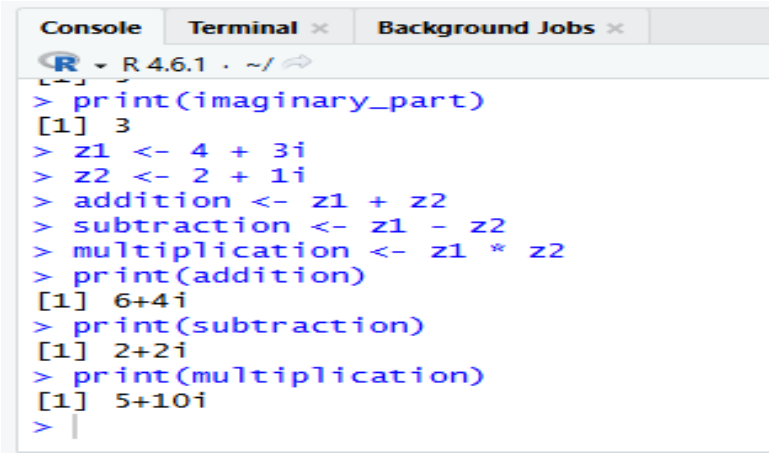
7. Create a complex number and display real and imaginary parts




```
[1] 8.4
> z <- 5 + 3i
> real_part <- Re(z)
> imaginary_part <- Im(z)
> print(z)
[1] 5+3i
> print(real_part)
[1] 5
> print(imaginary_part)
[1] 3
>
```

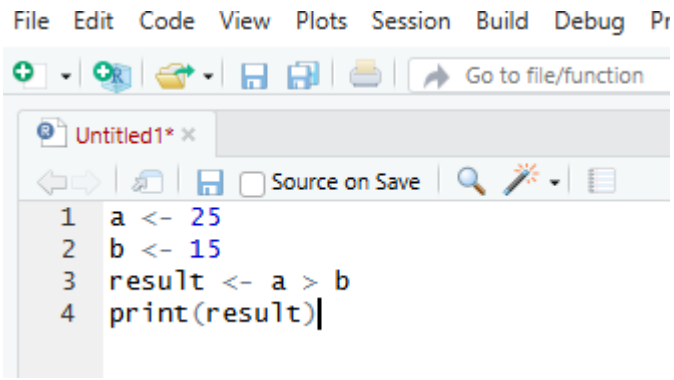
8. Create two complex numbers and perform operations





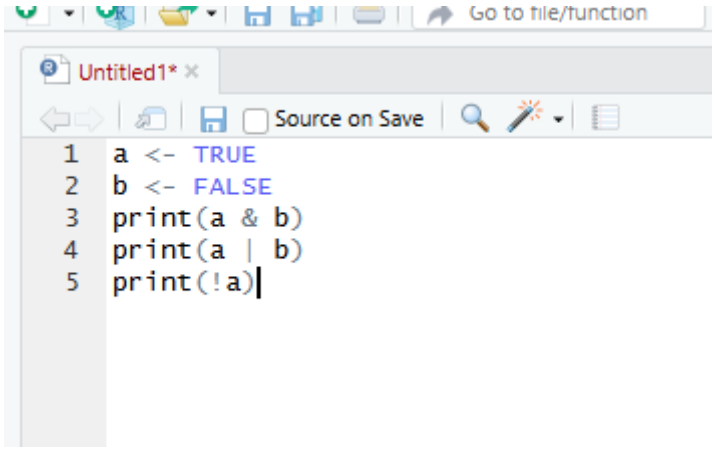
Logical Data Type

9. Compare two numbers and store result in logical variable




```
[1] 2+2i
> print(multiplication)
[1] 5+10i
> a <- 25
> b <- 15
> result <- a > b
> print(result)
[1] TRUE
>
```

10. Demonstrate logical operators &, |, and !




```
[1] TRUE
> a <- TRUE
> b <- FALSE
> print(a & b)
[1] FALSE
> print(a | b)
[1] TRUE
> print(!a)
[1] FALSE
>
```